

Epic 8: Building the Flask Web Application

Writing the Flask Backend

Flask is a lightweight Python web framework.

It is used to build web applications.

The backend handles user requests.

Routes connect URLs to Python functions.

The trained machine learning model is loaded.

User input is received through HTML forms.

The backend processes the input data.

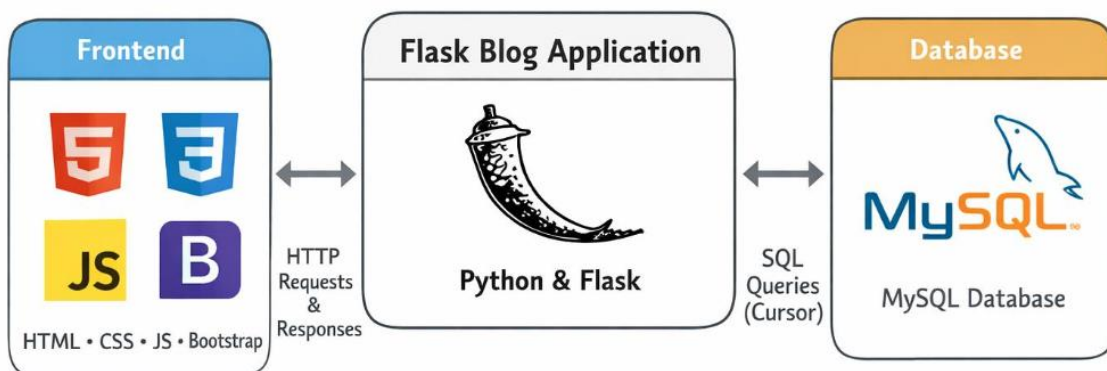
The model predicts the HDI value.

Prediction results are sent back to the webpage.

Flask simplifies deployment.

It integrates easily with machine learning.

This forms the application backend.



Running the Flask Application

The Flask server is started using Python.

The application runs on localhost.

The default port is 5000.

Users access the application through a browser.

The browser sends requests to Flask.

Flask processes every request.

The machine learning model generates predictions.

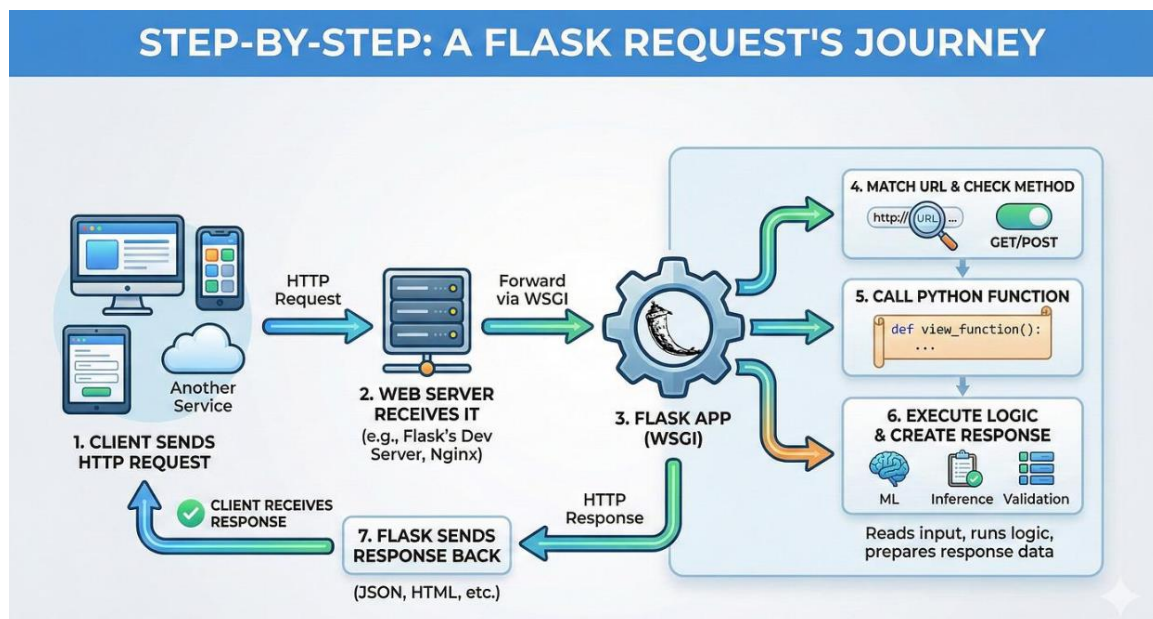
The output is displayed on the webpage.

Errors are shown in the terminal.

Debug mode helps developers.

Testing ensures the application works correctly.

The application is now ready for use.



Working of the Flask Web Application

The user enters input values.

The browser sends the data to Flask.

Flask validates the input.

The trained model predicts the result.

The prediction is returned to the frontend.

The webpage displays the output.

This process is fast and efficient.

Flask connects the frontend and backend.

The saved model is reused for predictions.

The system supports real-time prediction.

The HDI project becomes user-friendly.

This completes the Flask application workflow.

